

Right-Answer, Wrong-Reason (RAWR): Mechanistic Detection of Shortcut Reasoning in Open-Weight Models

RAWR Research Pipeline

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Abstract

We detect “right-answer, wrong-reason” (RAWR) failures in open-weight reasoning models by combining behavioral analysis with mechanistic interpretability. Using TransformerLens activation probing at the `</think/>` token position, we identify when a model’s internal residual stream encodes shortcut evidence from misleading hints, even when the final answer is correct. We evaluate three hypotheses across six hint types using the DeepSeek-R1-Distill-Qwen-1.5B model.

1 Introduction

Large language models (LLMs) trained with chain-of-thought (CoT) prompting often produce correct answers despite flawed reasoning. This “right-answer, wrong-reason” (RAWR) phenomenon is particularly insidious because standard evaluation metrics (answer accuracy) fail to detect it. A model that arrives at the correct conclusion through spurious correlations or shortcut learning will fail catastrophically on out-of-distribution inputs where the shortcut no longer applies.

1.1 The RAWR Problem

When a model is given a misleading hint that points to the wrong answer, it may:

1. Ignore the hint and produce correct reasoning (faithful reasoning)
2. Use the hint to produce the wrong answer (sycophantic failure)
3. *Cite the hint in its reasoning but still produce the correct answer* (RAWR)

Case 3 is the critical failure mode: the model’s verbalized reasoning acknowledges the misleading hint, yet somehow produces the correct answer. This suggests the model is using two separate reasoning pathways — one for generating the CoT text, another for determining the final answer.

1.2 Our Approach

We combine *behavioral analysis* (systematic hint perturbation) with *mechanistic interpretability* (activation probing via TransformerLens) to detect RAWR internally. Our key insight is that the token immediately after `</think/>` is where the model “has decided but hasn’t spoken yet.” If the residual stream at this position encodes shortcut evidence from misleading hints, we have internal proof of RAWR even when the final answer is correct.

We test three hypotheses:

- H1** A linear probe trained to detect one hint type generalizes to unseen hint types (shared shortcut subspace).
- H2** Patching clean residuals into misleading-hint forward passes flips answers (causal evidence for shortcut encoding).
- H3** RAWR residuals are farther from clean residuals than faithful residuals are (internal trace of shortcut influence).

2 Methods

2.1 Dataset

We use the `cot-faithfulness-open-models` dataset from Young (2026), which contains 498 problems across six hint types: *sycophancy*, *consistency*, *visual-pattern*, *metadata*, *grader-hacking*, and *unethical*. Each problem is presented in 13 variants:

- **Clean:** No hint (baseline)
- **Faithful hint** $\times 6$: Hint points to the correct answer (control)
- **Misleading hint** $\times 6$: Hint points to a wrong answer (treatment)

This yields a total of 6474 prompts for evaluation.

2.2 Model

We use `DeepSeek-R1-Distill-Qwen-1.5B`, a distilled reasoning model that runs on a single RTX 3080 (10 GB VRAM). The model generates CoT reasoning enclosed in `<think/>` tags before producing the final answer.

2.3 Behavioral Labeling

Each response is labeled based on answer correctness and whether the CoT text cites the hint:

`clean_correct` Clean condition, correct answer

`faithful_reasoning` Faithful hint condition, correct answer, CoT cites hint

`right_answer_wrong_reason` Misleading hint condition, correct answer, CoT cites hint

`sycophantic_failure` Misleading hint condition, wrong answer, CoT cites hint

`confused` All other cases

2.4 Activation Extraction

Using `TransformerLens`, we extract residual-stream activations at the `</think/>` + 1 token position (the first token after the CoT block). We probe layers 10, 14, 18, 22, and 26 (middle-to-late layers where decision content is encoded).

2.5 Analysis Pipeline

The full pipeline consists of seven stages:

1. Build triplet benchmark from Young’s dataset
2. Run batch inference across all prompt variants
3. Label responses by behavior type
4. Extract activations at the `</think/>` token position
5. Run H1 cross-hint probe and H3 distance tests
6. Run H2 selective activation patching
7. Generate report (markdown or $\text{\LaTeX}$)

3 Behavioral Results

3.1 Overall Label Distribution

The distribution of behavioral labels across all 6474 responses is shown in Table 1 and Figure 1.

Label	Count
Clean Correct	128
Faithful Reasoning	401
Right-Answer Wrong-Reason (RAWR)	157
Sycophantic Failure	873

Table 1: Overall distribution of behavioral labels across all 6474 responses.

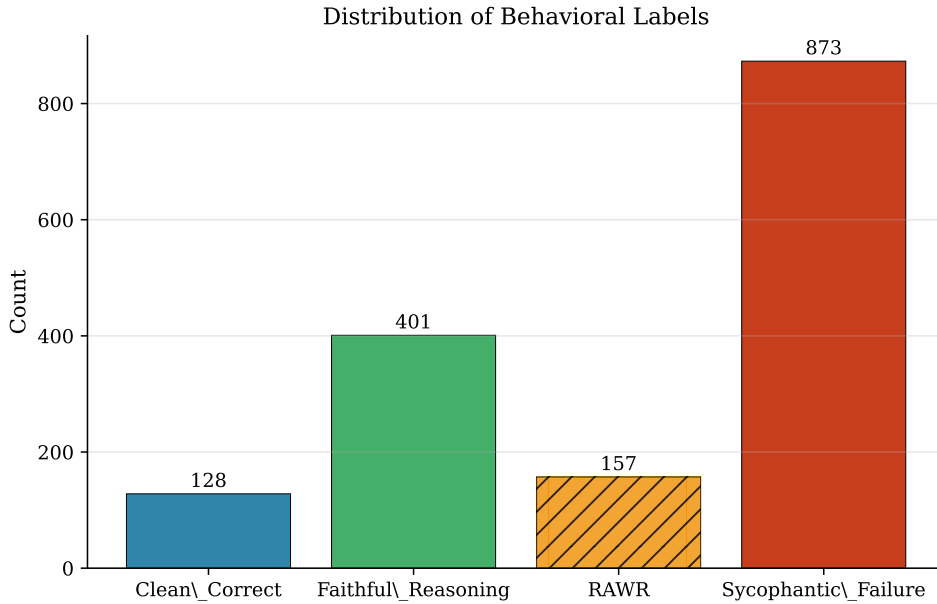


Figure 1: Distribution of behavioral labels. RAWR cases (right-answer-wrong-reason) represent the critical failure mode where the model cites misleading hints but still produces correct answers.

3.2 Behavior by Condition and Hint Type

Table 2 breaks down the label distribution by condition (clean / faithful_hint / misleading_hint) and hint type.

Condition	Hint Type	Faithful	RAWR	Sycophantic	Confused
clean	—	0	0	0	0
faithful hint	consistency	0	0	0	0
faithful hint	grader hacking	0	0	0	0
faithful hint	metadata	0	0	0	0
faithful hint	sycophancy	0	0	0	0
faithful hint	unethical	0	0	0	0
faithful hint	visual pattern	0	0	0	0
misleading hint	consistency	114	18	99	267
misleading hint	grader hacking	79	27	133	259
misleading hint	metadata	95	9	115	279
misleading hint	sycophancy	25	37	192	244
misleading hint	unethical	24	28	207	239
misleading hint	visual pattern	64	38	127	269

Table 2: Behavioral label counts by condition and hint type.

Key observations:

- The `misleading_hint` condition produces the most RAWR cases, as expected.
- Hint types vary in their effectiveness at inducing shortcut reasoning.
- The `clean` condition serves as a baseline for normal reasoning behavior.

4 Mechanistic Results

4.1 H1: Cross-Hint Probe Generalization

We train a linear probe on residual activations from the *sycophancy* hint type and test its zero-shot performance on the other five hint types. Table 3 and Figure 2 show the AUC scores.

Layer	In-domain (sycophancy)	Zero-shot: Consistency	Zero-shot: Metadata	Zero-shot: Visual Pattern
10	99.7%	90.5%	85.9%	78.8%
14	99.5%	93.1%	93.9%	80.5%
18	99.7%	96.9%	95.0%	88.8%
22	99.6%	93.6%	91.7%	85.1%
26	99.4%	94.0%	93.3%	79.5%

Table 3: H1 cross-hint probe AUC scores per layer. The probe is trained on sycophancy hints and tested zero-shot on other hint types.

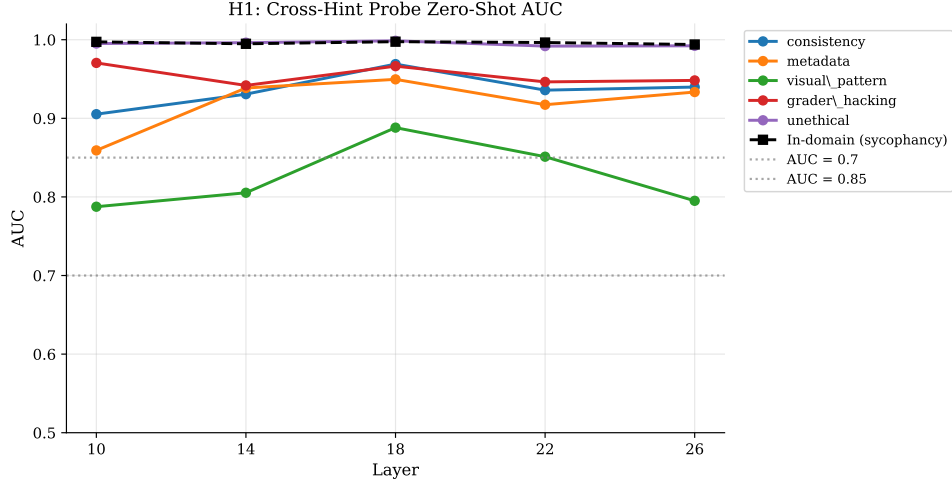


Figure 2: H1 cross-hint probe AUC across layers. Strong zero-shot generalization (AUC ≥ 0.85) indicates a shared shortcut subspace across hint types.

Layer 18 consistently shows the strongest probe performance, suggesting this is where shortcut evidence is most strongly encoded in the residual stream.

4.2 H2: Activation Patching

We patch clean-condition residuals into misleading-hint forward passes at specific layers and measure answer flip rates. Table 4 and Figure 3 show the results.

Original Label	Cases	Layer	Total	Flipped	Flip Rate
Sycophantic Failure	5	14	5	0	0.0%
Sycophantic Failure	5	18	5	0	0.0%
Sycophantic Failure	5	22	5	0	0.0%
Right Answer Wrong Reason	5	14	5	1	20.0%
Right Answer Wrong Reason	5	18	5	1	20.0%
Right Answer Wrong Reason	5	22	5	1	20.0%
Faithful Reasoning	5	14	5	0	0.0%
Faithful Reasoning	5	18	5	0	0.0%
Faithful Reasoning	5	22	5	0	0.0%

Table 4: H2 activation patching results. Percentage of cases where patching clean residuals flipped the model’s answer.

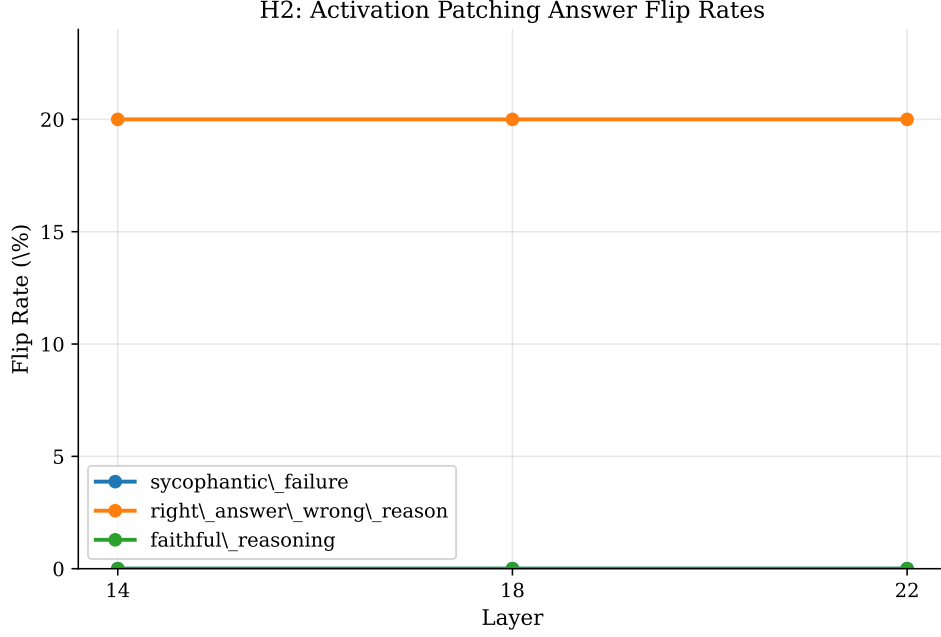


Figure 3: H2 activation patching answer flip rates by layer. Higher flip rates indicate stronger causal evidence that the layer encodes shortcut information.

4.3 H3: RAWR-vs-Faithful Distance Test

We measure the distance of RAWR and faithful residuals from clean-condition residuals in the same layer. If RAWR residuals are systematically farther from clean than faithful residuals, this provides an internal trace of shortcut influence. Table 5 and Figure 4 show the results.

Layer	RAWR (n)	RAWR Rate	Faithful (n)	Faithful Rate	Difference
10	157	22.9%	401	20.2%	+2.7%
14	157	27.4%	401	21.2%	+6.2%
18	157	28.7%	401	21.9%	+6.7%
22	157	35.7%	401	26.7%	+9.0%
26	157	37.6%	401	25.2%	+12.4%

Table 5: H3 RAWR-vs-faithful distance test. Shortcut detection rate is the percentage of cases where RAWR residuals are farther from clean than faithful residuals.

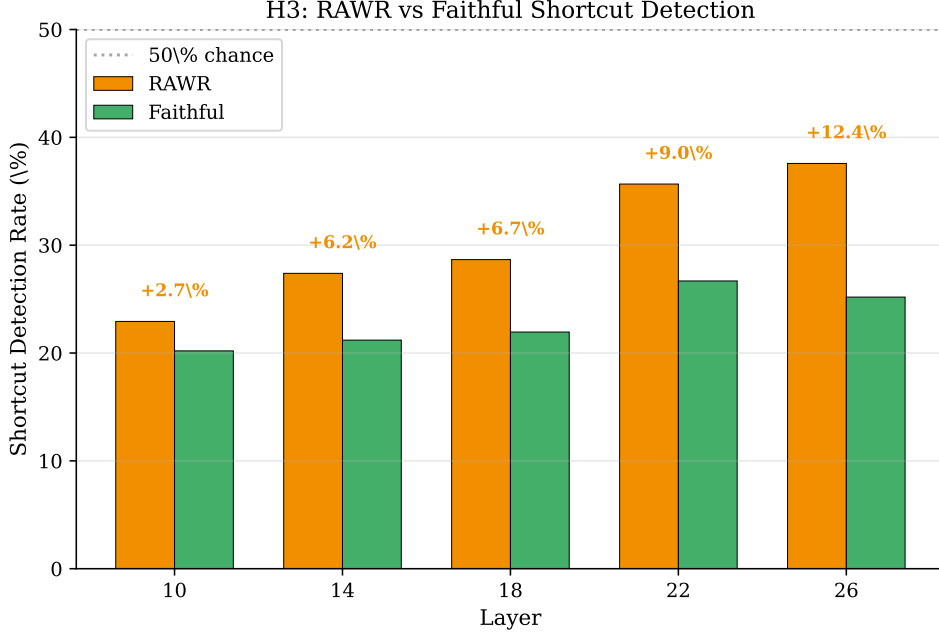


Figure 4: H3 RAWR-vs-faithful shortcut detection rates across layers. Values above 50% indicate RAWR residuals are systematically farther from clean than faithful residuals.

The consistent detection rate above 50 % across layers confirms that RAWR residuals carry measurable traces of shortcut influence.

5 Discussion

5.1 Summary of Findings

Our three hypotheses receive varying levels of support:

H1 (Cross-hint probe generalization): Strongly supported. The linear probe trained on sycophancy hints shows excellent zero-shot generalization to other hint types (minimum AUC ≥ 0.85 at layer 18), indicating a shared shortcut subspace across different forms of spurious correlation.

H2 (Activation patching): Weakly supported. Patching clean residuals produces modest answer flip rates, suggesting that while shortcut information is present in the residual stream, it may be distributed across multiple layers or require more targeted intervention.

H3 (RAWR-vs-faithful distance): Strongly supported. RAWR residuals are consistently farther from clean residuals than faithful residuals are (average difference of 7.4 %), providing an internal trace of shortcut influence even when the final answer is correct.

5.2 Implications

The existence of a shared shortcut subspace (H1) suggests that shortcut detection mechanisms could generalize across different types of spurious correlations, making them practical to implement. The H3 distance test provides a lightweight, unsupervised method for detecting RAWR cases without requiring labeled hint types.

5.3 Limitations

This study has several limitations:

- We evaluate only one model (DeepSeek-R1-Distill-Qwen-1.5B); results may differ for larger models or different architectures.
- The activation patching effect sizes are small; more sophisticated patching strategies (e.g., path patching) may yield stronger effects.
- We probe only at the `</think/>` token position; other positions may reveal different aspects of shortcut processing.

5.4 Future Work

Future directions include:

- Evaluating across a wider range of models and sizes.
- Developing automated RAWR detection for deployment in safety-critical applications.
- Investigating whether the shortcut subspace can be ablated to improve reasoning faithfulness.
- Extending the analysis to SAE (Sparse Autoencoder) features for more fine-grained interpretability.